

Coffee Shop Mid-Session Report

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1 Project Summary

The purpose of this project is to generate actionable market insights for the theoretical coffee chain *Grean Bean*.

Our current focus is exploratory data analysis of sales trends for April 2019, alongside identifying opportunities to increase revenue and reduce costs.

Key initiatives include developing recommender systems to boost sales, analyzing product performance, optimizing pastry inventory to minimize wastage, and leveraging stock data to anticipate future product demand.

2 Data Dictionary

We incorporate two data sources. The primary dataset is IBM’s Coffee Shop 11.1.3+ data, comprising nine CSV files: *2019 Sales Receipts*, *Dates*, *Customer*, *Generations*, *Pastry Inventory*, *Product*, *Sales Targets*, *Sales Outlet*, *Staff*. These are integrated into MySQL for advanced querying and analysis, capturing all sales activity for April 2019.

We also use the *Coffee Shop Sales* dataset from Maven Analytics, a hypothetical yet realistic sales dataset, as a secondary source.

3 DELTTAA Framework

We applied the DELTTAA methodology, introduced in *Competing on Analytics* by Tom Davenport, to assess and strengthen the coffee shop’s analytical maturity:

Data: The coffee shop recently adopted a new recordkeeping system, so no data predates April. Although April’s data is well-documented—covering sales, inventory, products, and customers—the limited historical scope currently places the business at Stage 2 for data maturity.

Enterprise: Data is applied across the organization, including customer records, receipts, wastage reports, and profitability metrics. This holistic use of data suggests the enterprise is approaching Stage 3 in analytical integration.

Leadership: While we lack direct insight into leadership, the move toward curated data processes indicates management is beginning to recognize the importance of analytics, indicative of Stage 3 aspirations.

Targets: Clear business goals have been defined, such as reducing pastry wastage, increasing customer net revenue, and forecasting seasonal demand, reflecting Stage 3 target-setting maturity.

Technologies: The company uses modern tools such as cloud-based databases, Python for analytics, and basic visualization platforms, but lacks more advanced technological investments.

Techniques (Analytical): Advanced methods such as clustering, predictive modeling, and optimization have been applied to customer segmentation, recommendation systems, and scheduling.

Analysts: The company currently lacks in-house analytics expertise, relying on external consultants, which suggests a Stage 1–2 level of analyst maturity.

4 FACE Framework

We applied the FACE framework to structure our analytics initiatives:

Frame: We identified three key business questions:

- What product recommendations can boost customer revenue?
- How can pastry wastage be minimized to reduce costs?
- How can stock data inform predictions of seasonal product demand?

Analytically Model: We developed recommender systems, optimization algorithms, and seasonal forecasting models. For recommendations, we explored collaborative filtering and apriori techniques to suggest complementary products. For optimization, we applied Markov processes and time series models to improve pastry production. For seasonal insights, we studied indicator products as predictors of broader demand patterns.

Communicate: Findings will be presented through interactive dashboards and training seminars for stakeholders.

Embed: Models are designed to integrate with existing systems and adapt dynamically as new data becomes available.

5 Pachinko Framework

The Pachinko Machine framework provides a structured way to approach analytics projects by guiding analysts through key decisions: defining the problem type, identifying its nature, and selecting appropriate modeling techniques. For *Green Bean*, we applied the framework as follows:

Step 1: Identify the problem, decision, or opportunity. We identified two ongoing operational challenges:

1. High product wastage, increasing costs and lowering profitability.

2. Missed cross-selling opportunities, leaving potential revenue untapped.

Step 2: Determine the modeling type. Both challenges are best addressed with predictive and prescriptive modeling rather than descriptive reporting alone.

Step 3: Assess the problem’s nature. These challenges are hybrid:

- Product demand is stochastic, influenced by time, weather, and customer behavior.
- Inventory and shelf space are deterministic, constrained by fixed resources.

Step 4: Select appropriate modeling techniques. We adopted complementary approaches:

- *For demand forecasting and waste reduction:* Time series forecasting and simulation (evaluative).
- *For product recommendations:* Association rule mining and collaborative filtering (probabilistic).
- *For inventory optimization:* Integer programming (normative).

Step 5: Implement and iterate. The resulting models are being embedded into point-of-sale and inventory systems to enable real-time decision-making and continuous improvement.

By applying the Pachinko Machine framework, we ensured each challenge was approached with a clear understanding of its complexity and aligned with the coffee shop’s strategic goals of cost efficiency and revenue growth.

6 Current Progress

We have explored several key metrics for the coffee chain. We created charts for net revenue per store per day (Figure 1).

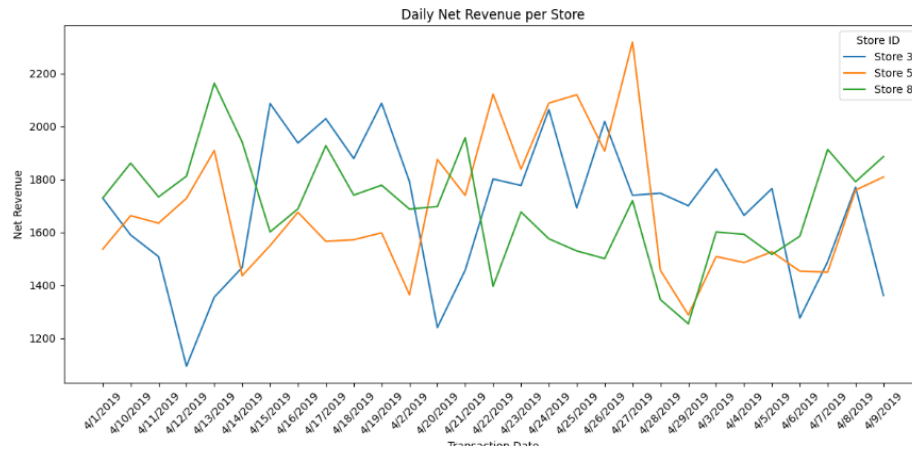


Figure 1: Daily net revenue per store

We aggregated net revenue over all of April to track total monthly net revenue by store (Figure 2).

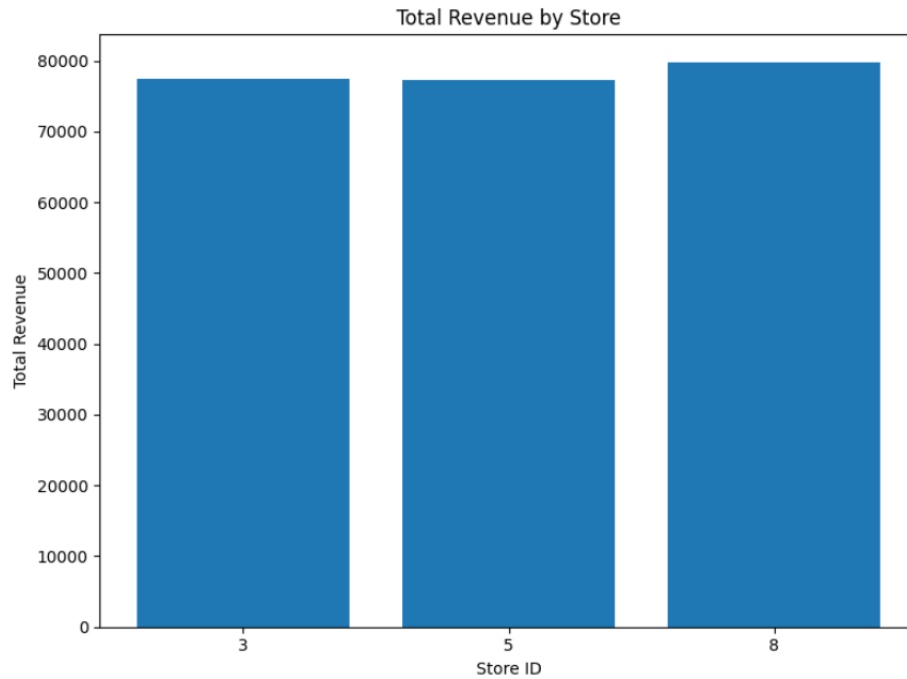


Figure 2: Monthly net revenue per store for April 2019

We generated pie charts for revenue by product per store. Here we show Store 3 (Figure 3).

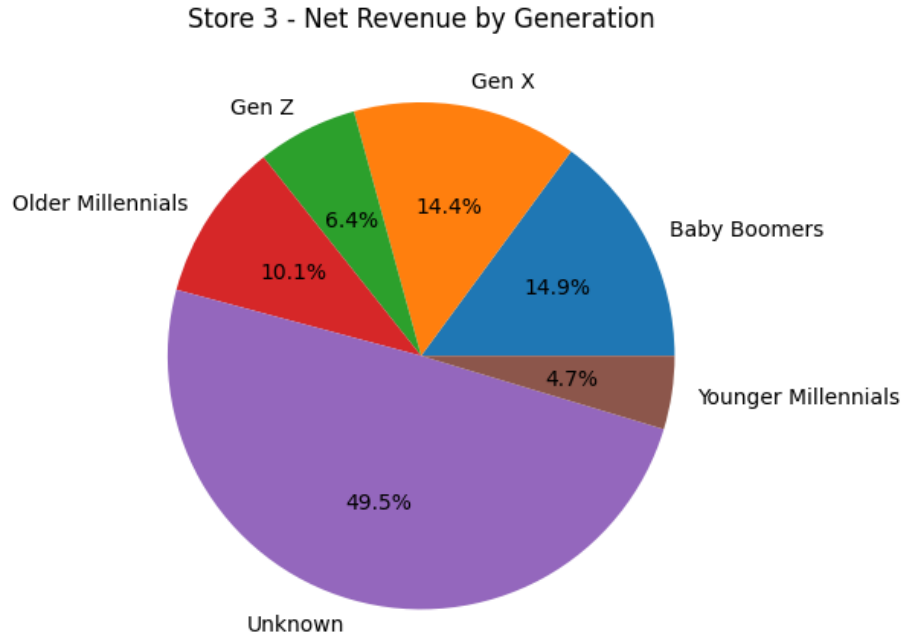


Figure 4: Net revenue by generation for Store 3

Other key findings indicate that 99% of orders have a net revenue under \$10; 81% are under \$5; and 40% are under \$3. This equates to 38,794 orders under \$10, 31,959 under \$5, and 15,516 under \$3. Hypothetically, if just 10% of the 15,516 smallest orders included an additional \$3 item, the shop would gain an extra \$4,654.80 in monthly revenue.

To support this, we began developing an apriori-based recommendation system to suggest low-cost add-ons to small orders. Results so far are mixed, but we continue to refine the approach.

7 Conclusion

While there is still much to be done, our findings this far suggest there is very clear room for improvement within the coffee company, and that with focused efforts we can uncover marketable insights to boost revenue.

8 Resources

- Kaggle. (n.d.). *Coffee Shop Sample Data 11.1.3+*. Retrieved from: <https://www.kaggle.com/datasets/ylchang/coffee-shop-sample-data-1113?select=201904+sales+reciepts>.

- Kaggle. (n.d.). *Coffee Sales Dataset*. Retrieved from: <https://www.kaggle.com/datasets/ahmedabbas757/sales>